DoRAemon's Secret Tool for an A in Deep Learning: Weight-Decomposed Fine-Tuning

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Cornell University CS 5782 / CS 4782 Intro to Deep Learning

Motivation & Paper Idea

Full fine-tuning large language models is expensive. **LoRA** reduces the cost by training low-rank adapter updates while freezing the base model, but the update can constrain how pretrained weights move.[1]

DoRA

- **DoRA** decomposes pretrained weights into **magnitude** and **direction**.[3]
- Direction receives a **LoRA**-style low-rank update; magnitude is learned separately.

Project goal

- Does **DoRA** improve commonsense accuracy over a same-scope **LoRA** baseline?
- Reproduce the LLaMA2-7B trend and test attention-only, MLP-only, and full-scope adapters.

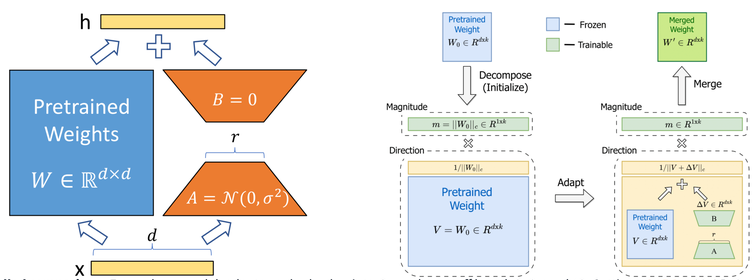
Methodology

LLaMA

Model, data, and metrics

- Base model: `meta-llama/Llama-2-7b-hf`.
- Training data: `commonsense_15k.json`; eval on eight commonsense benchmarks.
- Metrics: per-task and macro-average accuracy.

Adapter mechanics



Controlled comparison. Frozen base model, adapter-only checkpoints, 3 epochs, cutoff length 256, seed 42. Settings: **LoRA** r=32, LR=3e-4; rank-halved **DoRA** r=16, LR=2e-4.

Our Independent Experiment: Scope Ablation

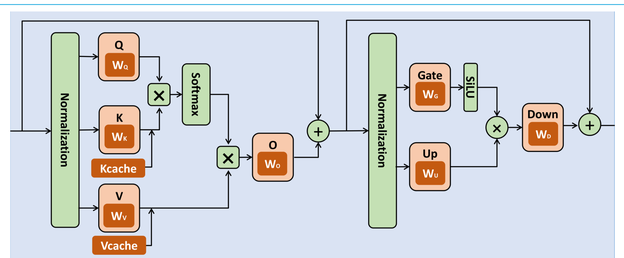


Figure 1. The ablation isolates adapter placement in attention projections, MLP projections, or both.

Scope	Target modules	Purpose
Full	`q,k,v,up,down` projections	Paper-level reproduction
Attention only	`q,k,v` projections	Test routing/context layers
MLP only	`up,down` projections	Test feed-forward layers

Headline Result

77.41%	79.20%	+1.79	6/1/1
Full **LoRA** macro	Full rank-halved **DoRA** macro	accuracy points	task wins/ties/losses

Figures 1 and 2 show that full-scope, rank-halved **DoRA** reproduced the paper's qualitative improvement over **LoRA** under our student-scale setup.

Macro Accuracy by Adapter Scope

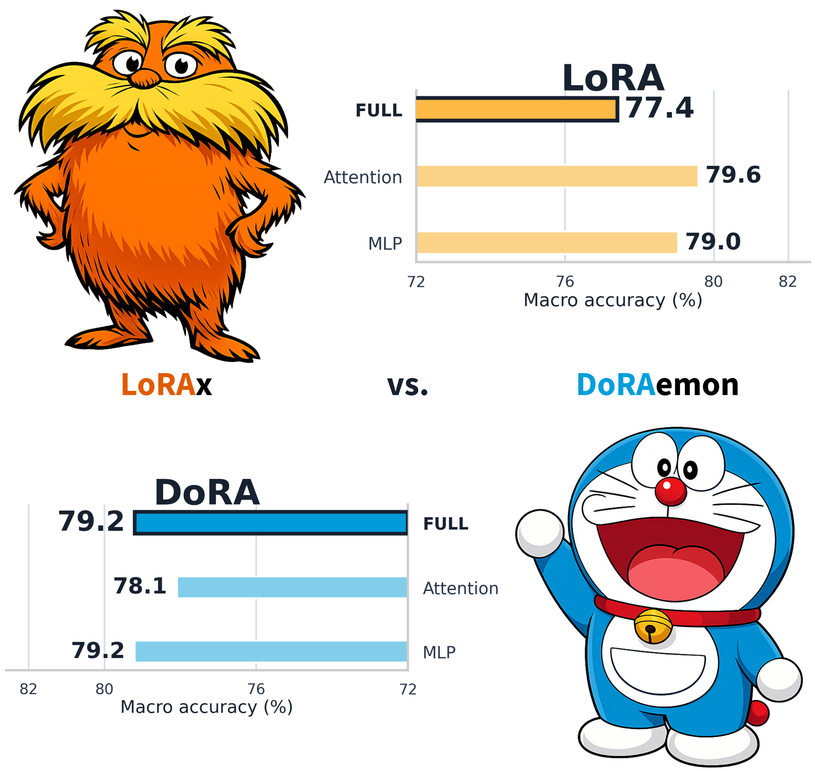


Figure 2. The full-scope bars are outlined: rank-halved **DoRA** improves macro accuracy from 77.4% to 79.2%.

HellaSwag: Largest DoRA Gain (+6.0 pts)

Task format. Choose the most plausible four-way continuation for a grounded activity description.

Full-scope DoRA vs. LoRA. 79.5% vs. 73.5% on HellaSwag, the largest single-task gain in our study.

Takeaway. The gain peaks when attention routing and MLP world knowledge adapt together; attention-only **DoRA** loses the benefit.

Official Reference Context

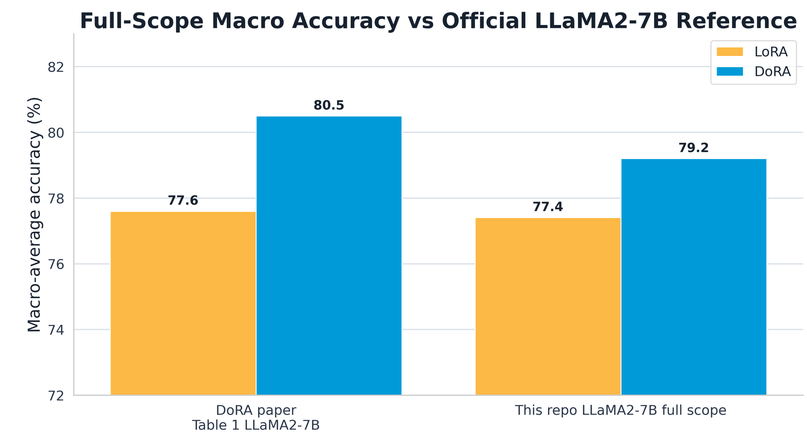


Figure 3. Our result matches the paper's direction: rank-halved **DoRA** improves over **LoRA**, but our 79.2% **DoRA** score remains below the official 80.5% reference.

Interpretation

- The main paper-level claim **partially reproduced**: full-scope **DoRA** beat full-scope **LoRA**.
- Scope ablations show the gain was **not automatic**: attention-only **DoRA** underperformed **LoRA** on 7/8 tasks; MLP-only **DoRA** was roughly tied.
- Key takeaway: **DoRA** appears most useful when attention and MLP transformations adapt together, not as a replacement for one module group.
- Key limitations: 15k subset instead of full data, seed 42 only, quantized Colab runs, and a local implementation rather than the exact official recipe.

Conclusion & Future Work

Takeaways

- Full-scope **DoRA** improved macro-average accuracy by **+1.79 points**.
- The reproduction supports the qualitative value of weight-decomposed low-rank adaptation.
- The ablation adds a caution: applying **DoRA** narrowly can remove the benefit.

Next steps

- Train on the full `commonsense_170k.json` dataset.
- Repeat all six conditions across multiple random seeds.
- Add parameter-count, GPU-memory, and wall-clock cost comparisons.

References

[1] Edward J. Hu et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[2] Hugging Face. Transformers documentation. `https://huggingface.co/docs/transformers`, 2026.

[3] Shih-Yang Liu et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. Oral presentation.

[4] NVIDIA Research. NVlabs/DoRA: Official implementation. `https://github.com/NVlabs/DoRA`, 2024.